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Chapter 1

Making a Map

Plotly is an open-source data visualization library for Python, R, and JavaScript. It allows for interactive plots, including geographical maps. In this brief example, we will learn how to create a simple annotated map using Plotly in Python.

To begin, you need to install Plotly. In Python, you can do this via pip:

```
pip install plotly
```

Once installed, you can create a map with annotations as follows:

```python
import plotly.graph_objects as go

fig = go.Figure(data=go.Scattergeo(
    lon = [-75.789],
    lat = [45.4215],
    text = ['Ottawa'],
    mode = 'text',
))

fig.update_layout(
    title_text = 'Annotated Map with Plotly',
    showlegend = False,
    geo = dict(
        scope = 'world',
        projection_type = 'azimuthal equal area',
        showland = True,
        landcolor = 'rgb(243, 243, 243)',
        countrycolor = 'rgb(204, 204, 204)',
    ),
)

fig.show()


```

This code creates a world map and places a text annotation at the geographic coordinates for Ottawa. Then, the plotly is opened in your browser using a localhost IP address. The ‘go.Scattergeo‘ function is used to define the geographical scatter plot (i.e., the annotation), while the ‘update_layout‘ function is used to define the appearance and the properties of the map itself.

In this example, you can replace the latitude, longitude, and text with the values corresponding to your desired location.

CHAPTER 2

Introduction to Discrete Probability

Before we begin talking about discrete probability, let's address the word *discrete*. It means "individually separate and distinct" as in "You can think of light as a continuous wave or as a blast of discrete particles."

When we are talking about probabilities, some problems deal with discrete quantities like "What is the probability that I will throw these three dice and the numbers that roll face up sum to 9?" There are also problems that deal with continuous properties like "What is the probability that the next bird to fly over my house will weigh between 97.2 and 98.1 grams?" In this module, we are going to focus on the probability problems that deal with discrete quantities.

Watch Khan Academy's Introduction to Probability at `https://youtu.be/uzkc-qNVoOk`.

Let's say that you have a cloth sack filled with 100 marbles; 99 are red and 1 is white. If you reach in without looking and pull out one marble, you will probably pull out a red one. We say that "There is a 1 in 100 chance that you would pull out a white marble." Or we can use percentages and say "There is a 1% chance that you will pull out a white marble." Or we can use decimals and say "There is a 0.01 probability that you will pull out a white marble." In probability, we often talk about the probability of certain events. "Pulling out a white marble" is an event, and we can give it a symbol like W. In equations, we use p to mean "the probability of". Thus, we can say "There is a 0.01 probability that you will pull out a white marble," which becomes the equation

$$p(W) = 0.01$$

2.1 The Probability of All Possibilities is 1.0

We know that you are either going to pull out a red marble or a white marble, so the probability of a white marble being pulled and the probability of a red marble being pulled must add up to 100%. Therefore, the odds of pulling out a red marble must be 99%, or 0.99. If we let the event "Pull out a red marble" be given by the symbol R, we can say:

$$p(R) = 1.0 - P(W) = 1.0 - 0.01 = 0.99$$

Now, let's say that you take a marble from the bag, then toss a coin. What is the probability that you will pull a white marble, then get heads on the coin? It is the product of the two probabilities: $0.01 \times 0.5 = 0.005$, so one-half of a one percent chance. Do the probabilities still sum to 1?

- White and Heads = $0.01 \times 0.5 = 0.005$
- White and Tails = $0.01 \times 0.5 = 0.005$
- Red and Heads = $0.99 \times 0.5 = 0.495$
- Red and Tails = $0.99 \times 0.5 = 0.495$

$0.005 + 0.005 + 0.495 + 0.495 = 1.000$ Yes, the probabilities of all the possibilities still add to 1. This is only true because *the events do not effect each other*; the events are ***independent***.

2.2 Independence and Dependence

In the last section, you learned that the probability of two events ("Pulling a red marble from the bag" and "Getting tails in a coin toss") is the product of the probability of each event: $0.99 \times 0.5 = 0.495$.

This is true if the two events are *independent*; that is, the outcome of one doesn't change the probability of the other. The example we gave is independent: It doesn't matter what ball you pull from the bag, the outcome of the coin toss will always be 50-50.

What is an example of two events that are not independent? The probability that a person is a professional basketball player and the probability that someone wears a shoe that is size 13 or larger is *not* independent; rather it is dependent. After all, height is an advantage in basketball, and most tall people also have large feet. So, if you know someone is a basketball player, they likely wear large shoes.

2.3 Correlation does not Imply Causation

It is often said that "Correlation does not imply causation". What does this mean? Breaking it down:

- Correlation is a state in which two things increase or decrease together; two variables are *linearly* related
- Imply means suggests, as in *there is evidence for*
- Causation means there is a cause and effect relationship in place between the two variables.

So saying correlation does not imply causation simply means: just because two variables are linearly connected, it cannot be said one variable causes the other.

Here are some examples:

- Ice Cream Sales, Mosquito bites, and Sunburn cases all increase during the summertime. However, none of these are causally connected.
- Dinosaurs were not able to read books. However, this does not mean that the inability to read caused dinosaur extinction.
- There was a large increase in toilet paper sales and people working from home in the years 2020-2021. Does this mean that one causes the other? Not necessarily. In this case a third factor C, the pandemic, caused a large global quarantine, leading to increased toilet paper usage and the inability to work in an office.

Exercise 1 Rolling Dice

If you have three six-sided dice to roll, what is the probability that you will roll a five on all three dice?

Working Space

Answer on Page 59

Exercise 2 Flipping Coins

If you have five coins to flip, what is the probability that at least one coin will come up heads?

Working Space

Answer on Page 59

2.4 Why 7 is the most likely sum of two dice

If you roll two dice, the sum will be 2, 12, or any number in between. It is very tempting to assume that the likelihood of any of those numbers is the same. In fact, the probability of a 2 is $\frac{1}{36} \approx 3\%$ and the probability of a 7 is $\frac{1}{6} \approx 17\%$. A 7 is six times more likely than a 12! Why?

When you roll the first die, there are six possibilities with equal probability. When you roll the second die, there are six possibilities with equal probability. So, there are a total of 36 possible events with equal probabilities: 1 then 1, 1 then 2, 2 then 1, 1 then 3, 3 then 1, and so on. Only one of these (1 then 1) adds to 2. However, six of these sum to 7: 1 then 6, 6 then 1, 2 then 5, 5 then 2, 3 then 4, and 4 then 3. So, a 7 is six times more likely than a 2.

Here is the complete table:

Sum							Count	Probability
2	1,1						1	1/36
3	1,2	2,1					2	1/18
4	1,3	2,2	3,1				3	1/12
5	1,4	2,3	3,2	4,1			4	1/9
6	1,5	2,4	3,3	4,2	5,1		5	5/36
7	1,6	2,5	3,4	4,3	5,2	6,1	6	1/6
8		2,6	3,5	4,4	5,3	6,2	5	5/36
9			3,6	4,5	5,4	6,3	4	1/9
10				4,6	5,5	6,4	3	1/22
11					5,6	6,5	2	1/18
12						6,6	1	1/36

If you don't believe these numbers, you could roll a pair of dice hundreds of times and make a histogram. However, it would be a tedious and time-consuming task — just the sort of thing that we make computers do for us.

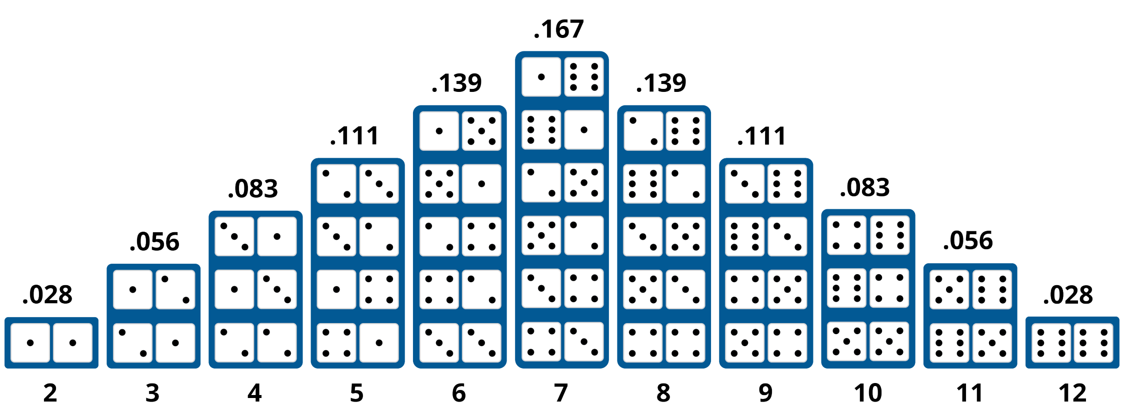

Figure 2.1: A histogram of probabilities of dice sums. This is referred to as a bell curve.

2.5 Random Numbers and Python

You are going to write a simulation of rolling dice in Python. To do this, you will need to generate a random sequence of numbers. The numbers will need to be in the range 1 to 6, and they will need to appear in the sequence with the same frequency. We say the sequence will follow *the uniform distribution*. In other words, the probability is uniformly distributed among the 6 possibilities.

Start python and try a few of the different ways to generate random numbers:

```
> python3
>>> import random
>>> random.random()  # Generates a random floating point number between 0 and 1
0.6840892758539989
>>> randrange(5)       # Generates an integer in the range 0 - 4
2
>>> x = ['Rock', 'Paper', 'Scissors']
>>> random.choice(x)   # Pick a random entry from the sequence
'Paper'
>>> x
['Rock', 'Paper', 'Scissors']
>>> random.shuffle(x)   # Shuffle the order of the sequence
>>> x
['Scissors', 'Paper', 'Rock']
>>> a = list(range(30))
>>> a
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,
  16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 29]
>>> random.sample(a, 10) # Return 10 randomly chosen items from the sequence
[8, 7, 20, 9, 25, 13, 23, 11, 14, 16]
```

Clearly, Python has many ways to do things that look random. However, here's the hidden truth: they aren't really random. The computer that you are using can't generate random data. Instead, it uses tricks to create data that look random; we call this *pseudorandom* data. Good pseudorandom algorithms are very important for cryptography and data security.

What if you want real random data? Some companies are using the decay of radioactive materials to generate real random data. You can pay to download it. For our purposes, Python's pseudorandom numbers are quite sufficient.

If we generate two random numbers in the range 1 through 6 and add them together, we will have simulated rolling a pair of dice. Like this:

```
>>> a = random.randrange(6) + 1
>>> b = random.randrange(6) + 1
>>> a + b
```

```
8
```

First, let's write a program that just rolls the dice 100 times and shows the result. Make a file `dice.py`:

```
import random

roll_count = 100

for i in range(roll_count):
    a = random.randrange(6) + 1
    b = random.randrange(6) + 1
    roll = a + b
    print(f"Toss {i}: {a} + {b} = {roll}")
```

When you run it, you should see something like:

```
> python3 dice.py
Toss 0: 6 + 6 = 12
Toss 1: 4 + 4 = 8
Toss 2: 4 + 2 = 6
Toss 3: 4 + 6 = 10
Toss 4: 4 + 4 = 8
...
Toss 98: 5 + 2 = 7
Toss 99: 5 + 2 = 7
```

Now we want to count occurrences of each possible outcome. Let's use an array of integers. We will start with an array of zeros. When we roll a 3, we will add 1 to item 3 in the array. (We can never roll a zero or a one, so those two entries will always be zero.)

```
import random

roll_count = 100

# Make an array containing 13 zeros
counts = [0] * 13

for i in range(roll_count):
    a = random.randrange(6) + 1
    b = random.randrange(6) + 1
    roll = a + b
    print(f"Toss {i}: {a} + {b} = {roll}")

    # Increment the count for roll
    counts[roll] += 1
```

```

print(f"Counts: {counts")}
```

When you run this, at the end you will see a count for each possible outcome:

```
...
Toss 98: 3 + 2 = 5
Toss 99: 6 + 1 = 7
Counts: [0, 0, 2, 6, 16, 11, 13, 14, 11, 11, 6, 9, 1]
```

What was the count that we expected? For example, we expected to see a 2 about once every 36 rolls, right? It might be nice to compare our count to what we expected. Add a few more lines, and we are going to increase the number of rolls. You will probably want to delete the line that prints each roll separately:

```
import random

# Can't ever be 0 or 1
p = [0.0, 0.0, 1/36, 1/18, 1/12, 1/9, 5/36, 1/6, 5/36, 1/9, 1/12, 1/18, 1/36]
roll_count = 1000

# Make an array containing 13 zeros
counts = [0] * 13

for i in range(roll_count):
    a = random.randrange(6) + 1
    b = random.randrange(6) + 1
    roll = a + b

    # Increment the count for roll
    counts[roll] += 1

for i in range(2,13):
    print(f"{i appeared {counts[i]} times, expected {p[i] * roll_count:.1f}")}
```

Now you should see something like:

```
2 appeared 39 times, expected 27.8
3 appeared 55 times, expected 55.6
4 appeared 84 times, expected 83.3
5 appeared 110 times, expected 111.1
6 appeared 160 times, expected 138.9
7 appeared 176 times, expected 166.7
8 appeared 124 times, expected 138.9
9 appeared 93 times, expected 111.1
```

```
10 appeared 87 times, expected 83.3
11 appeared 49 times, expected 55.6
12 appeared 23 times, expected 27.8
```

Whenever you are dealing with random numbers, the outcome will seldom be *exactly* what you expected. In this case, however, you should see that your predictions are pretty close.

2.5.1 Making a bar graph

A bar graph is a nice way to look at quantities like this. Let's make a bar graph that shows the actual count and the expected count:

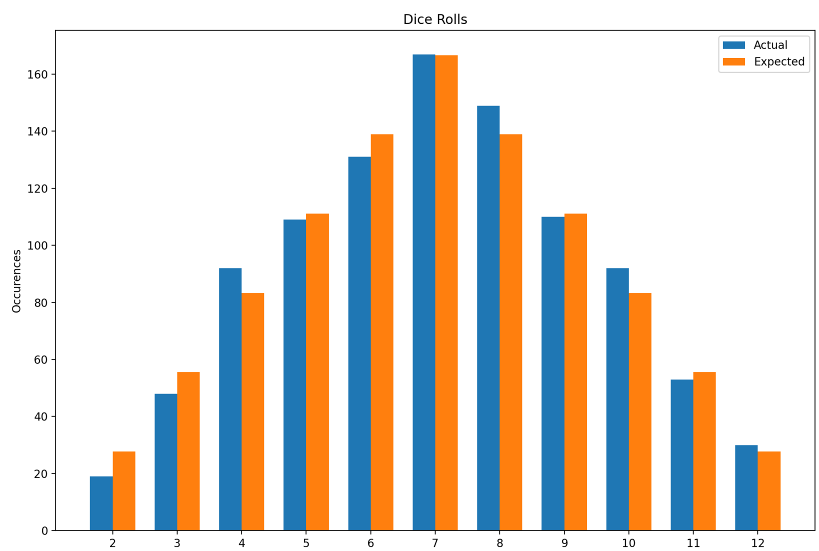

Figure 2.2: Dice roll histogram.

We need to describe the set of rectangles. To do this we will loop through each possible roll (2 - 12) and put data in four lists for each:

```
import random
import matplotlib.pyplot as plt

# Can't ever be 0 or 1
p = [0.0, 0.0, 1/36, 1/18, 1/12, 1/9, 5/36, 1/6, 5/36, 1/9, 1/12, 1/18, 1/36]
roll_count = 1000

```

```
# Make an array containing 13 zeros
counts = [0] * 13

for i in range(roll_count):
    a = random.randrange(6) + 1
    b = random.randrange(6) + 1
    roll = a + b

    # Increment the count for roll
    counts[roll] += 1

# Gather data for bar chart
bar_width = 0.35
expected = []
actual_starts = []
expected_starts = []
labels = []
actual = []
for i in range(2,13):
    expected.append(p[i] * roll_count)
    actual.append(counts[i])
    actual_starts.append(i - bar_width/2)
    expected_starts.append(i + bar_width/2)
    labels.append(i)

fig, ax = plt.subplots()

# Create the bars
ax.bar(actual_starts, actual, bar_width, label='Actual')
ax.bar(expected_starts, expected, bar_width, label='Expected')
ax.set_xticks(labels)

# Provide labels
ax.set_ylabel('Occurrences')
ax.set_title('Dice Rolls')
ax.legend()
plt.show()
```

Chapter 3

Beginning Combinatorics

Discrete probability problems often include some counting. For example, we figured out that there were 36 different ways the two dice could land, but all of them summed to some number 2 through 12. How many different ways could three eight-sided dice come up? We would need to count them, right? As the numbers get bigger, we will need some tricks so that we don't need to write them all down and count them one-by-one.

The branch of mathematics that focuses on tricks for counting is called *combinatorics*.

How can we be sure that there were 36 different configurations for the two 6-sided dice? The first die could have come up as any one of six numbers. For each of those, the second could have come up with any one of six numbers. Thus, the number of possibilities is $6 \times 6 = 36$.

How many different configurations for three 8-sided dice? $8 \times 8 \times 8 = 8^3 = 512$.

What about seven dice, each with 20 sides? There would be $20^7 = 1,280,000,000$ configurations. See, aren't you glad we don't need to write them all down?

Now, let's say that six people (Anne, Brock, Carl, Dev, Edgar, and Fred) are going to run a race. You have to make a plaque that says who won first place, who won second place, and who won third. If you want to get all the possible plaques created beforehand, and just pull the right one out as soon as the race ends, how many plaques would you need to get engraved?

Since you're making these plaques before the race even happens, you need one plaque for every possible outcome. So think of it as filling in three blanks: first place, second place, and third place. Any of the 6 people could fill the first blank. Whoever fills that blank can't also fill the second blank — a person can't come in both first *and* second — so only 5 people are left to fill the second blank. Likewise, only 4 people remain to fill the third blank, since two people have already been assigned to the first two spots. Multiplying these together, $6 \times 5 \times 4 = 120$, tells you exactly how many different first-second-third outcomes are possible — and thus how many plaques you'd need engraved.

What if the plaque includes all 6 places? Using the same reasoning, you'd fill in six blanks instead of three: $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$ plaques engraved. We use this process often enough that we gave it a name. We say "I need 6 factorial plaques engraved." When we write a *factorial*, we use an exclamation point:

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

We use the word "permutation" to mean a particular ordering. This rule says n items can be ordered in $n!$ ways. Thus, mathematicians actually say "If you have a list of n items then we can generate $n!$ different permutations of those items". We will discuss this further in the next chapter.

In Python, there is a `factorial` function in the math library:

```
> python3
>>> import math
>>> math.factorial(6)
720
```

Handy, right? Now you don't need to write a loop to calculate factorials.

Remember when we only wanted the first three names on the plaque? We can solve that problem using factorials:

$$6 \times 5 \times 4 = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = \frac{6!}{3!}$$

This formulation makes it easy to figure out on any calculator with a "!" button.

The rule on this is to fill m positions from n items. It can be done this many ways:

Permutations

For permutations where order is a factor, the formula for ${}^{n}P_{m}$ (n objects pick m) is

$$\frac{n!}{(n-m)!}$$

3.1 Choose

Let's say that there are 12 kids in a classroom and you need a team of four to clean the top of the desks. How many different possible teams are there? You know that if you were giving out four different positions (Like the race gave out 1st, 2nd, and 3rd), the answer would be $12 \times 11 \times 10 \times 9$ or $12!/(12-4)!$.

However, once we pick the 4 people, we don't care what order they are in, right? In this problem, the team "Anne, Brad, Carl, and Don" is the same as the team "Carl, Don, Brad, and Anne".

Thus, the quantity $12!/(12-4)!$ is many times too large, because it counts each permutation separately. To get the right number, we just divide this by the number of possible permutations for a group of four people: $4!$

That gets us our answer: How many different teams of four can be chosen from 12 people?

$$\frac{12!}{(12-4)!4!} = 495$$

Generally, this formula, referred to as the *Binomial coefficient*. In combinatorics, we use this quantity a lot, so we have given it a name: *choose*

Combinations

For combinations, where order doesn't matter, the formula for ${}^nC_k$ (n objects choose k) is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

We have also given it a notation. "12 choose 4" is written like this:

$$\binom{12}{4}$$

Python has the `math.comb` function:

```
> python3
>>> import math
>>> comb(12, 4)
495
```

We briefly saw this in the common polynomial products chapter. The binomial theorem says that the coefficients of the terms in the expansion of $(a+b)^n$ are given by the binomial coefficients $\binom{n}{k}$.

$$(a+b)^4 = 1a^4+4a^3b+6a^2b^2+4ab^3+1b^4 = \binom{4}{0}a^4+\binom{4}{1}a^3b+\binom{4}{2}a^2b^2+\binom{4}{3}ab^3+\binom{4}{4}b^4$$

3.2 Fractional Binomial Theorem

An issue arises when we have fractional exponents for this theorem. For example, what is $\sqrt{1+5x} = (1+5x)^{1/2}$? We can use the binomial theorem to expand this as follows:

$$(1+5x)^{1/2} = \binom{1/2}{0}1 + \binom{1/2}{1}5x + \binom{1/2}{2}(5x)^2 + \binom{1/2}{3}(5x)^3 + \ldots$$

Pascal's triangle only works when the exponent n is a non-negative integer. Here, the exponent is $\frac{1}{2}$, not a whole number. There's no "row $\frac{1}{2}$" of Pascal's triangle, so we need to extend the pattern.

This extension is called the ***general binomial theorem***. To make sense of $\binom{1/2}{k}$, we need a definition of the binomial coefficient that doesn't rely on factorials, since $(1/2)!$ doesn't make sense. Recall that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)(n-2)\cdots(n-k-1)}{k!}$$

Notice that the right-hand side only ever multiplies and divides — it never actually requires n to be a whole number! This gives us a definition that works for *any* real number n, including fractions and negative numbers:

Generalized Binomial Coefficient

For any real number n and non-negative integer k,

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}$$

When n is *not* a non-negative integer, like $n = \frac{1}{2}$, none of the factors $n, n-1, n-2, \ldots$ is ever zero, so the expansion never terminates. Instead of a finite row of Pascal's triangle, we get an infinite series.

General Binomial Theorem

Recall that for $|x| < 1$, the geometric series gives

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

Since $\frac{1}{1-x} = (1-x)^{-1}$, this looks like a binomial expansion where the exponent -1 is not a positive integer. Using the generalized binomial coefficient,

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}$$

For any real number n and $|x| < 1$,

$$(1+x)^n = \sum_{k=0}^{\infty} \binom{n}{k} x^k = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

this is called the *binomial series*, and it generalizes the binomial theorem to arbitrary real exponents. Setting $n = -1$ recovers the geometric series above.

The condition $|x| < 1$ matters here for convergence. When n was a positive integer, the expansion was just a polynomial, and polynomials are perfectly well-behaved for every value of x. But now that we have an infinite series, we need the terms to actually get smaller and smaller so the sum settles down to a finite value. If $|x| \geq 1$, the terms don't shrink, and the series doesn't converge to anything useful.

Let's compute the coefficients for $\binom{1/2}{k}$

$$\binom{1/2}{0} = 1$$
$$\binom{1/2}{1} = \frac{1}{2}$$
$$\binom{1/2}{2} = \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!} = \frac{\frac{1}{2}\cdot\left(-\frac{1}{2}\right)}{2} = -\frac{1}{8}$$
$$\binom{1/2}{3} = \frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!} = \frac{3/8}{6} = \frac{1}{16}$$

So,

$$(1+x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \cdots$$

Now we can go back to our original problem, $\sqrt{1+5x} = (1+5x)^{1/2}$. This is just $(1+x)^{1/2}$ with x replaced by $5x$, so we substitute $5x$ everywhere:

$$\begin{aligned}\sqrt{1+5x} &= 1 + \frac{1}{2}(5x) - \frac{1}{8}(5x)^2 + \frac{1}{16}(5x)^3 - \cdots \\ &= 1 + \frac{5}{2}x - \frac{25}{8}x^2 + \frac{125}{16}x^3 - \cdots\end{aligned}$$

This expansion is only valid for $|5x| < 1$, i.e. $|x| < \frac{1}{5}$.

Exercise 3 Fractional Binomial Expansion

1. Find the first four terms in the expansion of $(1+x)^{-1}$, in ascending powers of x. For what values of x is this expansion valid?
2. Find the first three terms in the expansion of $\sqrt{1-2x}$, in ascending powers of x.

Working Space

Answer on Page 59

3.3 Applications in Probability

Suppose you flip a fair coin 5 times. What's the probability that you get exactly 3 heads?

Note that there are only two options: Heads (H) or Tails (T). Thus, there are $2^5 = 32$ possible sequences of flips. Each sequence is equally likely, so the probability of any particular sequence is $\frac{1}{32}$. Only a certain number of sequences will have exactly 3 heads.

One way to get 3 heads is HHHTT. There's also HTHHT, THHHT, and many other orderings also give you 3 heads. Before we can find the probability, we need to know *how many* different sequences of 5 coin flips contain exactly 3 heads. As we have discussed, we can use the n-choose-k formula. There are

$$\binom{5}{3} = \frac{5!}{3!2!} = 10$$

such sequences. This is said "5 choose 3", or, more explicitly "get 3 successes in 5 trials".

Each individual sequence, like HHHTT, has probability $\left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2$ of occurring, since each flip is independent. Since there are 10 equally likely sequences that give exactly 3 heads, we add up 10 copies of that probability:

$$P(\text{exactly 3 heads}) = \binom{5}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 = 10 \cdot \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$$

This pattern of counting the arrangements with $\binom{n}{k}$, then multiplying by the probability of one arrangement works whenever you have repeated *independent* trials with only two possible outcomes each time (success or failure).

Exercise 4 Free Throws

A free throw shooter has a past record of making 60% of her free throws. Over the next 6 free throws, what is the probability that she makes exactly 4 of them?

Working Space

Answer on Page 59

This probability situation has been rightfully called the *binomial distribution*. This is a brief introduction to the binomial distribution, and we will discuss it in more detail in a later chapter.

Chapter 4

Permutations and Sorting

In the previous chapter, we talked about permutations. If you have a list of four letters, like $[a, b, c, d]$, you can rearrange them in 4! ways:

a,b,c,d	a,b,d,c	a, d, b, c	a, d, c, b	a, c, b, d	a, c, d, b
b,a,c,d	b,a,d,c	b, d, a, c	b, d, c, a	b, c, a, d	b, c, d, a
c,b,a,d	c,b,d,a	c, d, b, a	c, d, a, b	c, a, b, d	c, a, d, b
d,b,c,a	d,b,a,c	d, a, b, c	d, a, c, b	d, c, b, a	d, c, a, b

You can make Python generate all the permutations for you:

```python
from itertools import permutations
all_permutations = permutations(('a', 'b', 'c', 'd'))
for p in all_permutations:
    print(p)
```

4.1 Notation

How do we define or write down a single permutation? You could say something like "Swap the first and second items and swap the third and fourth items." However, that gets pretty difficult to read, so we usually write a permutation as two lines: the first line is before the permutation and the second line is after. Like this:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$$

We can also assign permutations to variables. For example, if we wanted the variable A to represent "swapping the first and second item", we would write this:

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix}$$

And if we wanted B to represent "swapping the third and fourth item", we would write:

$$B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix}$$

Now, we can *compose* permutations together. For example, we might say:

$$B \circ A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$$

In other words, if we have the list $[a, b, c, d]$ and we apply permutation A, followed by permutation B, we get $[b, a, d, c]$.

Important: Note that permutations are applied from right to left. $B \circ A$ means "Applying A and then B." Why does this matter? Permutations are not necessarily commutative. That is, if you have two permutations S and T, $S \circ T$ is not always the same as $T \circ S$.

Also, note that "don't change anything" is a permutation. We call it *the identity permutation*. If you have four items, the identity permutation would be written:

$$I = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

(We use a capital "I" for the identity.)

4.1.1 Challenge

Find an example of two permutations S and T, such that $S \circ T$ does not equal $T \circ S$.

4.2 Sorting in Python

One of the common forms of permutation in software is sorting. Sorting is putting data in a particular order. For example, in Python, if you had a list of numbers, you can sort it in ascending order like this:

```
my_grades = [92, 87, 76, 99, 91, 93]
grades_worst_to_best = sorted(my_grades)
```

Do you want to sort backwards?

```
my_grades = [92, 87, 76, 99, 91, 93]
grades_best_to_worst = sorted(my_grades, reverse=True)
```

Note that `sorted` makes a new list with the correct order. If you want to sort the array in place, you can use the `sort` method:

```
my_grades = [92, 87, 76, 99, 91, 93]
my_grades.sort(reverse=True)
```

4.3 Inverses

Think for a second about this permutation:

$$S = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$$

You could say this permutation shuffles a list. What is its inverse? That is, what is the permutation that unshuffles the items back to where they were originally?

$$S^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

In other words, the original moved an item in the first spot to the third spot; the inverse must move whatever was in the third spot back to the first spot.

(Notation note: In multiplication, $b \times b^{-1} = 1$, so we use "to the negative one" to indicate inverses in lots of places.)

Mechanically, how do you find the inverse? Flip the rows, then sort the columns using the top number:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \text{ flip } \rightarrow \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \text{ sort } \rightarrow \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

Let's say you have two permutations A and B. Permuting by B then A would look like this:

$$C = A \circ B$$

If you know A^{-1} and B^{-1}, what is C^{-1}? You would undo-A, then undo-B, so

$$C^{-1} = B^{-1} \circ A^{-1}$$

4.4 Cycles

Here is a permutation:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 1 & 3 \end{pmatrix}$$

When this is applied, whatever is at 1 gets moved to 2, 2 gets moved to 4, and 4 gets moved to 1. That is a *cycle*: $1 \to 2 \to 4$, then it goes back to 1. It involves three locations, so we say it is a *3-cycle*.

There is another cycle in this permutation: $3 \to 5$, then it goes back to 3.

Because these cycles share no members, we say the cycles are *disjoint*.

Every permutation can be broken down into a collection of disjoint cycles.

$$T = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 1 & 3 \end{pmatrix} = (1 \to 2 \to 4)(3 \to 5)$$

The first handy thing about this notation is that it makes it easy for us to describe the inverse. We just run the cycles backward:

$$T^{-1} = (4 \to 2 \to 1)(5 \to 3)$$

Starting with the list $[a, b, c, d, e]$, let's repeatedly apply the permutation T

Initial	a, b, c, d, e
T applied	d, a, e, b, c
$T \circ T$ applied	b, d, c, a, e
$T \circ T \circ T$ applied	a, b, e, d, c
$T \circ T \circ T \circ T$ applied	d, a, c, b, e
$T \circ T \circ T \circ T \circ T$ applied	b, d, e, a, c
$T \circ T \circ T \circ T \circ T \circ T$ applied	a, b, c, d, e

This permutation results in six combinations, then it loops back on itself. The number of combinations is the least common multiple of all the cycles. In this case, there is a 3-cycle and a 2-cycle. The least common multiple of 2 and 3 is 6.

4.5 Further Learning

Check out this leetcode problem which combines permutations and programming! https://leetcode.com/problems/permutations/description/

Chapter 5

Statistical Inferences for Proportions

5.1 Interval Estimation and the Confidence Interval

Suppose you had to guess your own weight, and then guess a classmate's weight. You would probably feel more confident about your own guess. You might say 140 pounds for both, but you would trust your own number a lot more than the one for your classmate. A point estimate only gives you a single number. It does not tell you how much to trust that number.

A point estimate is a single value that is calculated from a sample. It serves as our best guess for a population parameter. Although it can be useful, it is incomplete. Two point estimates can look identical on paper while carrying very different amounts of certainty behind them, like the way your weight guess and your classmate's guess might both be 140 pounds while being backed by very different amounts of information.

This is where the *confidence interval* comes in. A confidence interval gives a range of plausible values for the population parameter that is built around the point estimate. The *margin of error* is the "plus or minus" amount that sets the width of that range. It is driven by how confident we want to be and how much the estimate varies from sample to sample.

A wider interval is more likely to capture the true value, but it also tells you less. This tradeoff between confidence and precision runs through everything in this chapter.

5.2 Constructing a Confidence Interval for a Proportion

When we discussed confidence intervals, we saw that a wider interval gives us more certainty at the cost of being precise. That tradeoff is captured by a single number called the *critical value*, written z^*.

Let's say a city council wants to estimate the proportion of residents who support building a new network of bike lanes. Before collecting any data, they need to decide how confident they want to be in their eventual estimate, say 95%. That confidence level determines z^*.

A 95% confidence level leaves 5% split evenly between the two tails of the standard normal distribution, so 2.5% in each tail. We need the value z^* such that the area to its right is 0.025, which means the area to its left is 0.975. Using a standard normal table or a calculator's inverse normal function, $z^* \approx 1.96$.

If the city council instead wanted 99% confidence, would z^* be larger or smaller than 1.96?

Once we have z^*, we still need to decide how precise we want our estimate to be. An interval can technically be true and still be worthless. If the council's confidence interval for bike lane support turned out to be "somewhere between 5% and 95%," that would be almost useless for making a decision, even if it is a perfectly valid 95% confidence interval. The width of a useful interval, called the margin of error, depends on how large a sample the council collects. Larger samples produce narrower, more informative intervals for the same confidence level.

This is why sample size planning matters just as much as the confidence level. A researcher who wants a useful interval has to think ahead about how much data collecting that precision will require.

Computing in Excel

Computing in Python

5.3 Inference: Tests of Significance

Building a confidence interval answers "what is the population proportion likely to be?" But sometimes the question you actually care about is different: has something changed? *Statistical inference* is the process of using information from a sample to answer a specific question or make a decision about a population parameter, and a *test of significance* is the tool built for exactly this kind of question.

Chapter 6

Statistical Inferences for Means

In Statistical Inferences for Proportions, every question came down to a percentage: what proportion of trial users converted to paying customers, and whether that proportion changed after a redesign. Not every question reduces to a percentage.

Sometimes what you want to know is a genuine measurement, how long something takes, how much something weighs, how far something travels, on average.

A quantitative variable like this calls for the same two tools you already trust, a confidence interval to estimate a mean and a significance test to check a claim about one, but neither tool behaves the way it did for proportions. This is because sometimes you are missing crucial information: you almost never know the population's true standard deviation, only what your own sample can tell you. That gap forces a new distribution into the picture, which looks like the normal curve.

6.1 The t-Distribution and Conditions for Inference about a Mean

Suppose the streaming service now wants to know something different from Statistical Inferences for Proportions's conversion rate: how long, on average, a subscriber watches in a single sitting. A product analyst pulls a random sample of viewing sessions and calculates a sample mean watch time, $\bar{x}$. To turn that one number into a confidence interval or a significance test, you need the standard error of $\bar{x}$, and the standard error depends on the population standard deviation, σ. Here is the catch: nobody knows σ. The population's true spread in viewing habits is exactly the kind of thing you would need a census to pin down, and if you had a census, you would not need inference in the first place.

6.1.1 Meet the t-Distribution

The sample standard deviation s estimates σ, but substituting s changes $\frac{\bar{x}-\mu}{s/\sqrt{n}}$ from a normal distribution to a t-*distribution*.

The t-distribution is continuous, symmetric, and centered at 0, but shorter and thicker-tailed than the normal distribution. Its shape depends on *degrees of freedom*, abbreviated df. For a single mean, $df = n-1$. As n grows, the t-distribution converges to the standard normal distribution.

The critical value t^* replaces z^* and is read from a t-table using both df and the tail area.

For $n = 18$ $(df = 17)$, a 90 percent confidence level gives $t^* = 1.740$, versus $z^* = 1.645$ for a known-σ case. At $df = 100$, $t^* = 1.984$, much closer to z^*.

Exercise 5 Reading the t-Table

A logistics company samples $n = 25$ delivery times and wants to construct a 95 percent confidence interval for the true mean delivery time. State the degrees of freedom, and find t^*.

Working Space

Answer on Page 59

6.1.2 Conditions for Inference about a Mean

Two conditions must hold for a t-procedure to be valid.

Independence: the data come from a random sample or randomized experiment, and if sampling without replacement, the sample size is no more than 10 percent of the population.

Normal or large sample: the sampling distribution of $\bar{x}$ is approximately normal. Check this using a dotplot, stemplot, or normal probability plot for strong skew or outliers, or rely on the population being known normal, or $n \geq 30$.

Use t, not z, unless σ is explicitly given, a case that rarely occurs in practice or on the AP exam.

- Independence: random sample or randomized experiment, and $n \leq 10\%$ of the population when sampling without replacement.
- Normal or large sample: no strong skew or outliers in the data, the population is known to be normal, or $n \geq 30$.
- Default to a t-procedure. Only switch to z if σ is explicitly given.

Exercise 6 Spot the Error

A student is asked to construct a confidence interval for a population mean using a sample of $n = 22$ measurements. The student writes, "Since $n = 22$ is less than 30, I cannot use a t-interval. I need to collect a bigger sample first." Explain what is wrong with this reasoning.

Working Space

Answer on Page 60

6.1.3 When the Data Comes in Pairs

Some data comes in pairs: two measurements per subject, such as watch time before and after a recommendation-algorithm update for the same subscribers. The two measurements within a pair are not independent.

Data like this is called *matched pairs*. Subtract to get one difference per subject, $d_i = x_i - y_i$, reducing the data to a single sample of differences. The t-distribution, degrees of freedom, and the independence and normality conditions all apply directly to that sample of differences. Independence now refers to the pairs being independent of one another, not the two measurements within a pair.

Matched pairs is not the same as comparing two independent means: two independent means needs two separate random samples, while matched pairs needs one sample of paired subjects.

Exercise 7 Paired or Independent?

For each scenario, state whether the data described would be analyzed as matched pairs or as two independent samples, and justify your answer in one sentence.

(a) Twenty subscribers each have their nightly watch time recorded for one week before a recommendation algorithm update and one week after.

(b) A researcher randomly splits 200 subscribers into two groups, shows one group only comedy titles and the other only drama titles for a month, and compares the two groups' average watch time.

Working Space

Answer on Page 60

6.2 Constructing and Justifying a Confidence Interval for a Population Mean

You now have everything you need to actually build an interval. Every confidence interval takes the shape:

$$\text{point estimate} \pm \text{margin of error}$$

6.2.1 The Confidence Interval for a Population Mean

The point estimate for a population mean μ is the sample mean, $\bar{x}$. The margin of error is the critical value t^* times the standard error of $\bar{x}$, which is $\frac{s}{\sqrt{n}}$. Putting the two together,

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

is a $(1-\alpha)100\%$ confidence interval for μ, using t^* from a t-distribution with $n-1$ degrees of freedom.

Back to the streaming service. The analyst samples $n = 24$ viewing sessions and finds a sample mean watch time of $\bar{x} = 47.8$ minutes with a sample standard deviation of $s = 9.6$ minutes. A dotplot of the sample shows no strong skew or outliers, and the sessions were pulled at random from the full session log, well under 10 percent of all sessions logged that month. Both conditions are satisfied, so a t-interval is appropriate.

For a 95 percent confidence interval, $df = 23$ and $t^* = 2.069$. The standard error is $\frac{9.6}{\sqrt{24}} \approx 1.960$, so the margin of error is $2.069 \times 1.960 \approx 4.05$ minutes. The interval is

$$47.8 \pm 4.05 \Rightarrow (43.75,\ 51.85)$$

so the analyst estimates the true mean session watch time to be between 43.75 and 51.85 minutes.

Exercise 8 Building a t-Interval

A city transportation office samples $n = 16$ commute times and finds $\bar{x} = 28.4$ minutes and $s = 6.2$ minutes. Conditions for inference are satisfied. Construct a 98 percent confidence interval for the true mean commute time.

Working Space

Answer on Page 60

6.2.2 Interpreting and Justifying the Interval

An interpretation references two things: the confidence level and the population. For the streaming service example: the analyst is 95 percent confident that the true mean watch time per session, across all sessions, falls between 43.75 and 51.85 minutes.

Do not say there is a 95 percent probability that μ falls in this interval. Once calculated, μ either is or is not inside it. The 95 percent describes the method: if the sampling process were repeated many times, about 95 percent of the resulting intervals would capture the true μ.

A confidence interval can justify a claim by checking whether the claimed value falls inside or outside it. If the product team's target is an average watch time of at least 45 minutes, the interval $(43.75, 51.85)$ dips below 45, so it does not provide convincing evidence that the true mean is at least 45 minutes.

For a fixed confidence level, interval width shrinks as n grows, roughly proportional to $\frac{1}{\sqrt{n}}$: quadrupling n from 24 to 96 shrinks the margin of error from about 4.05 to about 1.95 minutes, since $\sqrt{4} = 2$. For a fixed sample, raising the confidence level widens the interval.

Computing in Excel

Excel does not have a single built-in function that hands back a finished confidence interval, but two functions cover everything once you have n, $\bar{x}$, and s.

If you have the raw session lengths in a column, say `A2:A25`, get the three summary values first:

- `=AVERAGE(A2:A25)` for $\bar{x}$
- `=STDEV.S(A2:A25)` for s
- `=COUNT(A2:A25)` for n

Suppose those land in cells `B1` ($n = 24$), `B2` ($\bar{x} = 47.8$), and `B3` ($s = 9.6$), with the confidence level in `B4` (`0.95`).

The most direct route is `CONFIDENCE.T`, which returns the margin of error by itself:

- `=CONFIDENCE.T(1-B4, B3, B1)` returns 4.054, the margin of error.

Then the interval bounds are just `=B2-C1` and `=B2+C1`, where `C1` holds that margin of error.

If you would rather see each piece of the formula spelled out, build it manually instead:

- `=B1-1` for df
- `=T.INV.2T(1-B4, C1)` for t^*, where `C1` is the df cell (`T.INV.2T` takes the two-tailed probability, so use $1 - 0.95 = 0.05$ directly, not 0.025)
- `=B3/SQRT(B1)` for the standard error
- `=C2*C3` for the margin of error, where `C2` and `C3` hold t^* and the standard error

Both routes return the same margin of error, 4.054, and the same interval, $(43.75, 51.85)$. The manual version is worth building at least once, since it makes visible exactly where the t-distribution and the sample size are doing their work; `CONFIDENCE.T` is what you would reach for afterward, once you trust the formula.

Exercise 9 A Matched-Pairs Interval

A call center tests a new training program on $n = 12$ employees, recording each employee's average call-handling time before and after training. The sample of differences (before minus after) has $\bar{d} = -3.2$ minutes and $s_d = 4.5$ minutes. Conditions for inference on the differences are satisfied. Construct a 90 percent confidence interval for μ_d, and state what it suggests about the training program.

Working Space

Answer on Page 60

Exercise 10 Spot the Error

A student reports the streaming-service interval $(43.75, 51.85)$ and writes, "There is a 95 percent probability that the true mean watch time is between 43.75 and 51.85 minutes." Explain what is wrong with this interpretation, and rewrite it correctly.

Working Space

Answer on Page 61

6.3 Setting Up and Carrying Out a Test for a Population Mean

6.3.1 Setting Up the Test

A test for a population mean starts the same way a test for a proportion did: state hypotheses about the parameter, never about a sample statistic. The null hypothesis takes the form $H_0 : \mu = \mu_0$, where μ_0 is whatever value is being challenged. The alternative hypothesis is $H_a : \mu < \mu_0$, $H_a : \mu > \mu_0$, or $H_a : \mu \neq \mu_0$, depending on which direction the question points.

Choosing the method and verifying conditions works exactly as it did in the last section: check independence, check that the population is normal or the sample is large enough and free of strong skew or outliers, and default to a t-test unless σ is somehow known. If the data arrives as matched pairs, subtract first, then run everything that follows on the sample of differences, testing $H_0 : \mu_d = 0$ against whichever alternative fits the question.

6.3.2 Carrying Out the Test

The test statistic for a population mean is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

with $n - 1$ degrees of freedom. Find the p-value from the t-distribution, matching the tail(s) to H_a, and compare to α. Reject H_0 if p-value $< \alpha$.

A streaming service wants to know whether average session watch time has increased above the old baseline of 42 minutes after a redesign. Sample: $n = 30$, $\bar{x} = 44.6$ minutes, $s = 8.2$ minutes. Conditions are satisfied.

$$H_0 : \mu = 42 \qquad H_a : \mu > 42$$

$df = 29$, standard error $= \dfrac{8.2}{\sqrt{30}} \approx 1.497$, so

$$t = \frac{44.6 - 42}{1.497} \approx 1.737$$

p-value ≈ 0.0465. Since $0.0465 < 0.05$, reject H_0: sufficient evidence that mean session watch time increased above 42 minutes. A smaller p-value is stronger evidence against H_0 than one just under α.

Matched-pairs example: call-handling time, $\bar{d} = -3.2$ minutes, $s_d = 4.5$ minutes, $n = 12$. Testing $H_0 : \mu_d = 0$ against $H_a : \mu_d < 0$ gives $df = 11$, $t \approx -2.463$, p-value ≈ 0.0157.

At $\alpha = 0.05$, reject H_0: sufficient evidence the training program decreased average call-handling time. A confidence interval and a significance test on the same data will always agree.

Exercise 11 Setting Up and Testing a Claim

A bakery's recipe is supposed to produce loaves with a mean weight of 500 grams. A random sample of $n = 16$ loaves has $\bar{x} = 498.2$ grams and $s = 6.4$ grams, with no strong skew or outliers in the sample. At $\alpha = 0.05$, is there sufficient evidence that the true mean loaf weight differs from 500 grams?

Working Space

Answer on Page 61

Exercise 12 A Matched-Pairs Test

A wellness program measures resting heart rate for $n = 15$ participants before and after an eight-week exercise regimen. The sample of differences (before minus after) has $\bar{d} = 4.8$ beats per minute and $s_d = 6.1$ beats per minute. Conditions for inference on the differences are satisfied. At $\alpha = 0.05$, is there sufficient evidence that the regimen decreased mean resting heart rate?

Working Space

Answer on Page 61

Exercise 13 Spot the Error

A student sets up a test with the hypotheses $H_0 : \bar{x} = 500$ and $H_a : \bar{x} \neq 500$. Explain what is wrong with this setup.

Working Space

Answer on Page 61

6.4 Confidence Intervals for a Difference in Two Means

Every example so far has looked at one population at a time. Just as often, the real question is a comparison: not "what is the mean watch time," but "is the mean watch time different between two groups."

6.4.1 Two Independent Samples

Comparing average watch time between free-tier and paid-tier subscribers, two disjoint groups, means the samples are independent. Let μ_1, μ_2 be the true means. The parameter of interest is $\mu_1 - \mu_2$, point estimate $\bar{x}_1 - \bar{x}_2$.

$$\mathrm{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Degrees of freedom come from the Welch-Satterthwaite approximation: between $\min(n_1 - 1, n_2 - 1)$ and $n_1 + n_2 - 2$.

Conditions: both samples independently and randomly selected, each $n \leq 10\%$ of its population if sampling without replacement, and each sample free of strong skew/outliers or $n \geq 30$.

Sample: free-tier $n_1 = 20$, $\bar{x}_1 = 35.2$, $s_1 = 8.4$; paid-tier $n_2 = 18$, $\bar{x}_2 = 44.6$, $s_2 = 10.1$.

Conditions satisfied. For a 95 percent CI,

$$SE = \sqrt{\frac{8.4^2}{20} + \frac{10.1^2}{18}} \approx 3.032, \quad df \approx 33, \quad t^* \approx 2.035$$

Margin of error ≈ 6.17 minutes, point estimate $= -9.4$ minutes:

$$-9.4 \pm 6.17 \Rightarrow (-15.57,\ -3.23)$$

95 percent confident the true difference (free-tier minus paid-tier) is between -15.57 and -3.23 minutes. Since the entire interval is negative, paid-tier subscribers watch more per session on average.

Exercise 14 Building a Two-Sample Interval

A tech reviewer compares battery life between two phone models. Model A: $n_1 = 15$, $\bar{x}_1 = 11.2$ hours, $s_1 = 1.8$ hours. Model B: $n_2 = 17$, $\bar{x}_2 = 9.8$ hours, $s_2 = 2.1$ hours. The two samples are independent, and conditions for inference are satisfied. Construct a 90 percent confidence interval for the true difference in mean battery life, Model A minus Model B. Technology gives $df \approx 29$.

Working Space

Answer on Page 61

Exercise 15 Justifying a Claim

Using the battery-life interval from the previous exercise, does the data provide convincing evidence that Model A's true mean battery life exceeds Model B's by more than one hour? Explain using the interval.

Working Space

Answer on Page 62

Exercise 16 Spot the Error

A student compares two independent samples with visibly different spreads, one sample's dotplot is tightly clustered, the other is spread out much wider, and writes, ”Since both populations are approximately normal, I'll pool the two sample standard deviations into one estimate of σ and use that in my confidence interval.” Explain what is wrong with this reasoning for an AP Statistics response.

Working Space

Answer on Page 62

6.5 Significance Tests for a Difference in Two Means, and Selecting the Right Procedure

6.5.1 Setting Up and Carrying Out the Test

TTesting a claim about $\mu_1 - \mu_2$ follows the same four-step process, usually $H_0 : \mu_1 - \mu_2 = 0$, with $H_a : \mu_1 - \mu_2 < 0, > 0$, or $\neq 0$.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$$

using the same Welch-approximated df and conditions from Section 4.

Free-tier versus paid-tier, same samples as Section 4: $n_1 = 20$, $\bar{x}_1 = 35.2$, $s_1 = 8.4$; $n_2 = 18$, $\bar{x}_2 = 44.6$, $s_2 = 10.1$. Testing whether paid-tier watches more on average:

$$H_0 : \mu_1 - \mu_2 = 0 \qquad H_a : \mu_1 - \mu_2 < 0$$

$SE = 3.032$ (from Section 4), so

$$t = \frac{35.2 - 44.6}{3.032} \approx -3.100$$

$df \approx 33$, one-sided p-value ≈ 0.00196. Reject H_0: convincing evidence paid-tier subscribers watch more per session on average. A test and a confidence interval on the same data will always agree.

Exercise 17 Running a Two-Sample Test

Two store locations track customer wait times. Store 1: $n_1 = 22$, $\bar{x}_1 = 8.3$ minutes, $s_1 = 2.1$ minutes. Store 2: $n_2 = 25$, $\bar{x}_2 = 9.6$ minutes, $s_2 = 2.8$ minutes. The samples are independent, and conditions for inference are satisfied. Test, at $\alpha = 0.05$, whether Store 1's mean wait time is shorter. Technology gives $df \approx 44$.

Working Space

Answer on Page 62

6.5.2 Selecting the Right Procedure

By this point you have four inference procedures for means and, from the last chapter, four more for proportions. The hardest part of a real problem is often not the arithmetic, it is recognizing which of those eight procedures actually fits. Figure 1 lays out the same three questions, in order, that will get you there every time.

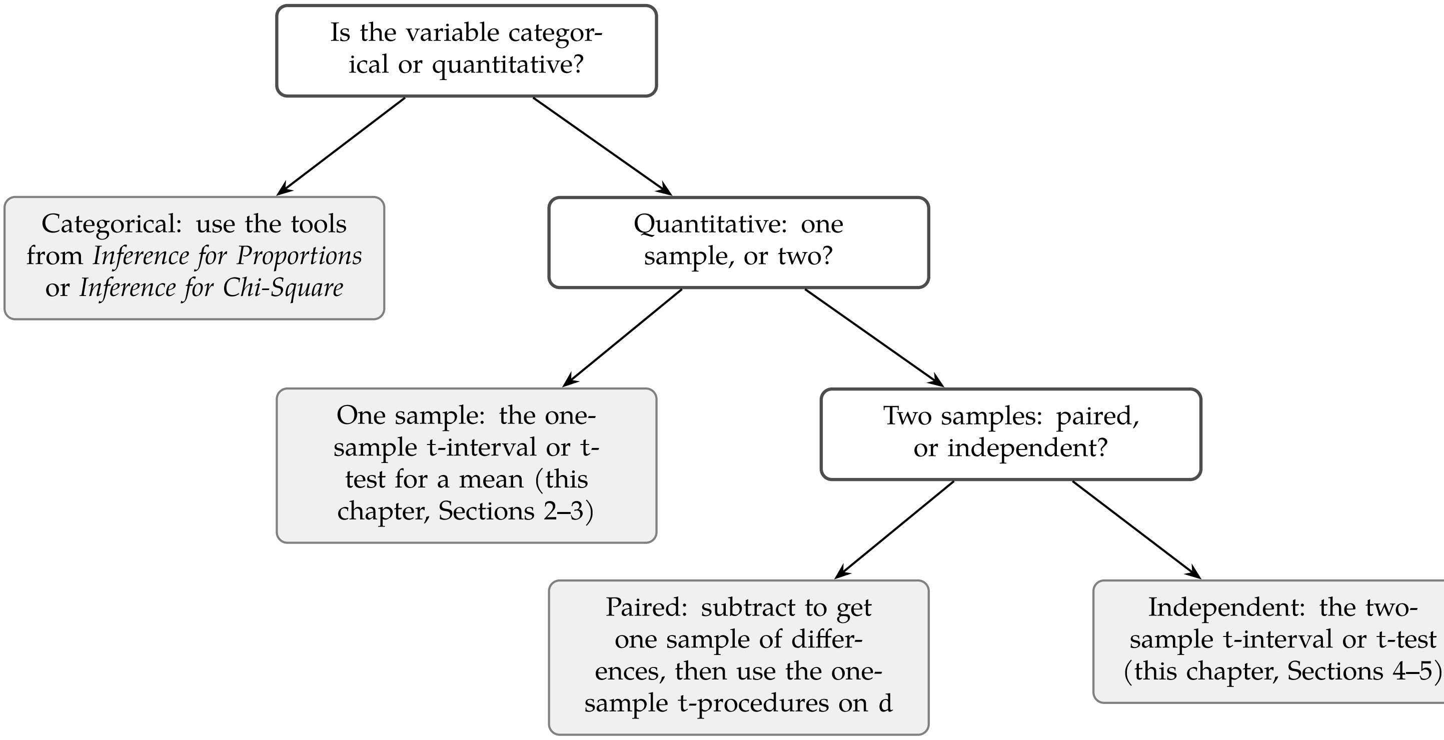

Figure 6.1: Selecting an inference procedure for a single quantitative or categorical variable

Categorical or quantitative is usually obvious from the variable itself. Paired or independent is worth slowing down for: does each measurement in one group have a natural partner in the other, or could the two groups' labels be shuffled without anything becoming meaningless? Same subscriber measured twice: shuffling is meaningless, so paired. Two disjoint groups: shuffling changes nothing, so independent.

6.5.3 Common Errors

Type I error, Type II error, and power all carry over from the last chapter without any changes to their definitions, just a change in what the parameter is. A Type I error here means rejecting $H_0 : \mu_1 = \mu_2$ when the two population means really are equal, concluding the streaming service's paid tier watches more when it actually does not. A Type II error means failing to reject H_0 when there really is a difference, missing a real effect. Power is still the probability of correctly detecting a real difference when one exists, and it still increases with a larger sample size, a larger true difference, or a higher significance level, exactly as it did for proportions.

Exercise 18 Naming the Error

In the store wait-time test from the previous exercise, the data led to rejecting H_0 in favor of Store 1 having a shorter mean wait time. Suppose, unknown to the researcher, the two stores' true mean wait times are actually equal. What type of error, if any, occurred? Explain.

Working Space

Answer on Page 62

Exercise 19 Choosing the Procedure

For each scenario, state which of the four procedures from Figure 1 applies, quantitative one-sample, quantitative paired, quantitative independent-samples, or categorical.

(a) A gym wants to know whether the average weight lifted by 30 randomly selected members changed after switching to a new equipment brand, measuring the same members' max lift before and after the switch.

(b) A researcher wants to compare the proportion of two different email subject lines that get opened.

(c) A nutritionist samples 40 adults at random and wants to estimate the true mean daily sugar intake in the population.

Working Space

Answer on Page 63

That decision flowchart will keep earning its keep. The next chapter turns to a different kind of categorical question, one with more than two possible outcomes at once, and the first thing you will need to decide, again, is which procedure actually fits the shape of the data in front of you.

Chapter 7

Statistical Inferences for Slopes

7.1 The Regression Model and Why We Need Inference for the Slope

A sleep researcher pulls data from a sample of 8 teenagers: daily screen time, in hours, and how many hours they slept that night. She plots the points, fits a line, and finds a slope of $b_1 = -0.16$: for every extra hour of screen time, sleep drops by about ten minutes. Does this mean the effect is real across all teenagers, or could a sample this size show a slope like this by chance?

Recall from the Numerical Methods chapter that the least-squares line $\hat{y} = b_0 + b_1 x$ describes your sample: descriptive statistics, not inference.

A sample slope b_1 estimates an unknown population slope β_1, the same way $\hat{p}$ estimates p and $\bar{x}$ estimates μ. Statisticians describe the true relationship between two quantitative variables using a *population regression model*:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

β_0 is the population y-intercept, β_1 is the population slope, and ε is random error: the part of Y not explained by X. β_0 and β_1 are parameters, describing the entire population; b_0 and b_1 are statistics, calculated from your sample, used to estimate them.

The rest of this chapter reasons from b_1, the number your sample gives you, to β_1, the value you actually care about, using the same confidence interval and significance test tools you've already built for proportions and means.

Estimating the Slope and Intercept

The formulas for b_0 and b_1:

$$b_1 = r\frac{s_y}{s_x} \qquad \text{and} \qquad b_0 = \bar{y} - b_1\bar{x}$$

where r is the sample correlation coefficient and s_x, s_y are the sample standard deviations of X and Y. If you are working from raw sums instead of r, s_x, and s_y, an algebraically equivalent formula is

$$b_1 = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

Both formulas give the same b_1. You can use whichever matches the summary statistics you have.

The residual for any data point is still $e = y - \hat{y}$, where $\hat{y} = b_0 + b_1x$ is the value your line predicts. The standard deviation of all the residuals in your sample, s_e, tells you the typical size of a prediction miss. You will use s_e heavily, because it is the key in measuring how much b_1 itself might vary from sample to sample.

Following the sleep researcher's study, she samples 8 teenagers at random and records their daily screen time (X, in hours) and their sleep duration that night (Y, in hours):

Screen time (hr)	2	3	4	5	6	7	8	9
Sleep duration (hr)	7.85	8.05	7.40	7.55	6.90	7.35	6.70	6.95

For this sample, $\bar{x} = 5.5$, $\bar{y} = 7.34$, $s_x = 2.45$, $s_y = 0.47$, and $r = -0.854$. That gives

$$b_1 = (-0.854)\frac{0.47}{2.45} = -0.165 \qquad b_0 = 7.34 - (-0.165)(5.5) = 8.25$$

so $\hat{y} = 8.25 - 0.165x$. The negative slope matches what the researcher suspected: more screen time is associated with less sleep, at least in this sample. The question this chapter answers is whether that association is strong enough and consistent enough, to conclude something about β_1 for teenagers in general, or whether a slope this far from zero is a believable result of sampling variation alone.

- $Y = \beta_0 + \beta_1 X + \varepsilon$ is the population regression model; β_0 and β_1 are unknown parameters, and ε is random error.
- b_0 and b_1 are sample statistics, calculated from your data, that estimate β_0 and β_1.
- $b_1 = r\frac{s_y}{s_x}$ and $b_0 = \bar{y} - b_1\bar{x}$, with an equivalent raw-sum formula available for b_1.
- The residual $e = y - \hat{y}$ and its standard deviation s_e measure how far actual data points fall from the line.

Exercise 20 Identifying the Model

A horticulturist records the amount of fertilizer applied (in grams per plant) and the resulting plant height (in centimeters) for a random sample of tomato plants.

1. Identify X and Y in this study.
2. Write the population regression model for this relationship, using correct notation.
3. In context, what would it mean if $\beta_1 = 0$?
4. Explain why β_1 is not something the horticulturist can ever know exactly, even after collecting data from every plant she owns.

Working Space

Answer on Page 63

Computing in Excel

Enter the sleep study data from earlier in this section into two columns: screen time in cells A2 through A9, and sleep duration in cells B2 through B9, with headers in row 1.

1. In an empty cell, enter `=SLOPE(B2:B9,A2:A9)`. Excel returns -0.1649, matching $b_1 = -0.165$ from the hand calculation (Excel simply keeps more decimal places).
2. In another cell, enter `=INTERCEPT(B2:B9,A2:A9)`. Excel returns 8.2506, matching $b_0 = 8.25$.
3. To check the correlation behind the calculation, enter `=CORREL(A2:A9,B2:B9)`. Excel returns -0.8536, matching $r = -0.854$.
4. To find s_x and s_y directly instead of computing them by hand, use `=STDEV.S(A2:A9)` and `=STDEV.S(B2:B9)`. These return 2.4495 and 0.4732.

Watch the argument order: for both `SLOPE` and `INTERCEPT`, the known y-values are listed first, then the known x-values. Reversing the two ranges is a common mistake, and it will silently hand you the wrong sign or a nonsense intercept rather than an error message.

7.2 The Sampling Distribution of b_1 and the Confidence Interval for the Slope

Suppose the sleep researcher ran her study again with a brand new sample of 8 teenagers. A new sample of 8 teenagers would give a different fitted line and a different b_1, the same way $\hat{p}$ and $\bar{x}$ vary from sample to sample. That variation is the *sampling distribution* of b_1: the distribution of values you'd get from repeated random samples of the same size. Under the conditions for inference, this distribution is approximately normal, centered at β_1, with standard deviation σ_{b_1}.

σ_{b_1} depends on σ_ε, the population standard deviation of the errors, which is unknown, so you estimate it with s_{b_1}, the *standard error of the slope*. Substituting s_{b_1} for σ_{b_1} turns

$$t = \frac{b_1 - \beta_1}{s_{b_1}}$$

from normal into a t-distribution with $df = n - 2$, the same trade as with means: estimated standard error in, precision out, t-distribution in place of normal.

Unlike proportions and means, you will almost never compute s_{b_1} by hand: it depends on s_e, sample size, and the spread of your x-values, and doing that from scratch is tedious enough that no exam or workplace expects it. Instead, s_{b_1} is read directly from a calculator's regression screen or software printout, in a column usually labeled "StDev" or "SE Coef." Recognizing and interpreting that column is the actual skill.

Constructing the Confidence Interval

With s_{b_1} in hand, a confidence interval for β_1 takes the same shape as every confidence interval you have built so far: statistic plus or minus a margin of error.

$$\text{Margin of Error} = t^* s_{b_1} \qquad \text{Confidence Interval: } b_1 \pm t^* s_{b_1}$$

where t^* is the critical value from a t-distribution with $df = n - 2$, chosen to match your desired confidence level.

Let's return to the sleep study. A statistical package fits the regression line to the 8 data points and reports the following output:

Predictor	Coef	StDev	T	P
Constant	8.2506	0.2448	33.70	0.000
Screen time	−0.1649	0.0411	−4.01	0.007

with $n = 8$, so $df = n - 2 = 6$. To build a 90% confidence interval for β_1, look up $t^* = t_{0.05}(6) = 1.943$. Then

$$\text{Margin of Error} = (1.943)(0.0411) = 0.0798$$

$$\text{Confidence Interval: } -0.1649 \pm 0.0798 = (-0.2447, -0.0850)$$

We are 90% confident that, for teenagers in general, each additional hour of daily screen time is associated with a change in average sleep duration of between −0.245 and −0.085 hours. Notice that this entire interval is negative. Since it does not contain 0, every plausible value of β_1 in this interval represents some negative relationship between screen time and sleep, which is a first hint that the effect the researcher found in her sample may reflect something real in the population, and not just sampling noise. Sections 3 and 4 make that hint precise, first by checking whether this procedure was even valid to use, and then by testing the claim directly.

- b_1 varies from sample to sample; its sampling distribution is centered at β_1 under the conditions for inference.
- The standard error s_{b_1} is read directly from calculator or computer regression output; it is rarely computed by hand.
- $t = \dfrac{b_1 - \beta_1}{s_{b_1}}$ follows a t-distribution with $df = n - 2$.
- Confidence interval: $b_1 \pm t^* s_{b_1}$, with t^* taken from the t-distribution at $df = n - 2$.
- If a confidence interval for β_1 does not contain 0, that is an early signal, not yet a conclusion, that the population slope may be nonzero.

Exercise 21 Reading Computer Output

A researcher studies the relationship between weekly study hours (X) and the improvement in exam score, in points (Y), for a random sample of $n = 12$ students. Regression output gives $b_1 = 2.35$ and $s_{b_1} = 0.68$.

1. Find the degrees of freedom for this problem.
2. Construct a 95% confidence interval for β_1. ($t^*_{0.025}(10) = 2.228$.)
3. Interpret the interval in context.

Working Space

Answer on Page ??

Computing in Excel

The regression output table above, including s_{b_1}, can be generated directly from the raw data rather than typed in by hand. Using the same screen time and sleep duration columns from Section 1 (A2:A9 and B2:B9):

1. Select a blank 2×2 range, say D2:E3. Type `=LINEST(B2:B9,A2:A9,TRUE,TRUE)` and confirm with Ctrl+Shift+Enter (in newer versions of Excel with dynamic arrays, a plain Enter is enough). This fills all four cells at once:

$b_1 = -0.1649$	$b_0 = 8.2506$
$s_{b_1} = 0.0411$	$s_{b_0} = 0.2448$

   The top row gives the coefficients, in reverse order from how you might expect (slope first, then intercept); the row directly below gives each one's standard error.

2. For the critical value, enter `=T.INV.2T(0.10,6)` in an empty cell. The first argument is $1 -$ confidence level, and the second is $df = n - 2$. Excel returns 1.9432, matching $t^*_{0.05}(6)$ from the table.

3. Margin of error: reference the two cells you already have, `=D3*[t* cell]`, giving 0.0798.

4. Confidence interval bounds: `=D2-[margin cell]` and `=D2+[margin cell]`, giving -0.2447 and -0.0850.

`LINEST` is the same idea as `SLOPE` and `INTERCEPT` from Section 1, but it returns the standard errors in the same step, so you never have to hunt for s_{b_1} separately. The argument order is the same gotcha as before: known y's first, known x's second.

7.3 Conditions for Inference on the Slope

The confidence interval you just built in Section 2 assumed that it was valid to use a t-procedure at all. Every inference method in this book comes with conditions attached, and the slope is no exception.

You have already met four of these five conditions for proportions and for means.

- **Linearity**: the true relationship between X and Y is actually linear, not curved.
- **Independence**: individual observations do not influence one another; if sampling without replacement, the population is at least 10 times the sample size.
- **Normality**: for any fixed value of X, the residuals are approximately normally distributed.
- **Equal variance**: the spread of the residuals is roughly the same across all values of X.
- **Random**: the data come from a random sample or a randomized experiment.

These are remembered by the acronym LINER. Random and Independence are conditions you check from the design of the study itself, the same way you always have. Linearity, Normality, and Equal variance are different: you check all three using the same tool, a *residual plot*, which graphs each residual against its corresponding x-value.

Checking the Conditions: The Sleep Study

Return to the sleep researcher's 8 teenagers. Before trusting the confidence interval from Section 2, check all five conditions.

Random. The researcher selected her 8 teenagers using a random sampling method, so this condition is satisfied by the design of the study.

Independence. Teenagers in general form a population far larger than 8, so sampling 8 of them without replacement does not meaningfully affect the independence of their screen time or sleep duration from one another. The 10% condition is easily satisfied.

Linearity, Normality, and Equal variance. These three come from the residuals themselves.

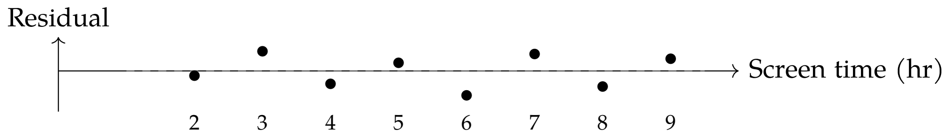

Figure 7.1: Residual plot for the sleep study regression.

The residuals in Figure 7.1 bounce above and below 0 with no visible curve, so linearity is reasonable: a straight line, rather than some curved model, is a sensible description of the relationship. The band of residuals also stays roughly the same width across the range of screen time values, from 2 hours to 9 hours, with no obvious funnel shape, so equal variance is reasonable as well.

Figure 7.2: Dotplot of the eight residuals from the sleep study regression.

The dotplot in Figure 7.2 shows no strong skew and no outliers among the eight residuals, so it is reasonable to proceed as though normality holds. With only $n = 8$ observations, though, this check is genuinely limited; a small sample simply does not give you enough information to rule out non-normality with much confidence. The honest conclusion is not "the residuals are normal," but "nothing in this small sample contradicts normality," which is a weaker, more defensible claim, and one that is worth stating explicitly on an exam or in a report.

With all five conditions reasonably satisfied, the confidence interval built in Section 2, and the hypothesis test coming in Section 4, both rest on solid ground.

- The five conditions for slope inference: Random, Linearity, Independence, Normality, Equal variance (LINER).
- Random and Independence come from the design of the study.
- Linearity, Normality, and Equal variance are checked using a residual plot (and, for normality specifically, a dotplot, boxplot, or normal probability plot of the residuals).
- A small sample size limits how confidently you can assess normality; conclusions should be stated as "nothing contradicts" rather than "confirms."

Exercise 22 Diagnosing a Residual Plot

A researcher studies the relationship between a car's weight and its fuel efficiency. The residual plot shows points that fan out widely for heavy cars but cluster tightly near 0 for light cars.

1. Which condition does this pattern call into question?
2. Explain, in your own words, why this pattern is a problem for the validity of a confidence interval or test built from this data.

Working Space

Answer on Page 63

7.4 Testing a Claim About the Slope

The confidence interval in Section 2 hinted that the true slope β_1 is not 0, since $(-0.2447, -0.0850)$ never crosses it. A hint is not a conclusion. To make the claim precise, and to attach an actual probability to the strength of the evidence, you need a significance test.

Step 1: State the Hypotheses

The sleep researcher's original question was directional: does more screen time predict *less* sleep, not just some unspecified difference. That points to a one-sided alternative.

$$H_0 : \beta_1 = 0 \qquad H_a : \beta_1 < 0$$

H_0 says screen time has no linear relationship with sleep duration in the population of teenagers; H_a says the relationship is negative. Use $\alpha = 0.05$.

Step 2: Name the Procedure and Verify Conditions

Use a one-sample t-test for the slope of a regression line, with $\mathrm{df} = n - 2$. Section 3 already checked all five conditions, Random, Linearity, Independence, Normality, and Equal variance, against this exact data set, and found no serious violations. Since the same data set is being reused here, that check still applies; there is no need to redo it.

Step 3: Calculate the Test Statistic and P-value

Recall the regression output from Section 2:

Predictor	Coef	StDev	T	P
Constant	8.2506	0.2448	33.70	0.000
Screen time	−0.1649	0.0411	−4.01	0.007

with $n = 8$, so $\mathrm{df} = 6$. The test statistic is

$$t = \frac{b_1 - 0}{s_{b_1}} = \frac{-0.1649}{0.0411} = -4.01$$

matching the T column exactly, since that column is always $\frac{\text{Coef}}{\text{StDev}}$.

The P column, 0.007, is a two-sided p-value: it treats $H_a : \beta_1 \neq 0$ as the alternative, whether or not that is the alternative you actually care about. Since our alternative here is one-sided, that number needs one adjustment before you can use it.

Converting a two-sided p-value to a one-sided p-value:

1. Check whether the sign of t matches the direction claimed in H_a. Positive t matches $H_a : \beta_1 > 0$; negative t matches $H_a : \beta_1 < 0$.
2. If the sign matches, divide the reported two-sided p-value by 2.
3. If the sign does not match, the data point the opposite way from your claim. The correct one-sided p-value is actually greater than 0.5, no matter how small the two-sided number looks, and there is no evidence for H_a.

Here $t = -4.01$ is negative, matching the direction of $H_a : \beta_1 < 0$, so the sign check passes. The one-sided p-value is

$$p\text{-value} = \frac{0.007}{2} = 0.0035$$

Step 4: State a Conclusion

Because $p = 0.0035 < \alpha = 0.05$, reject H_0. There is convincing statistical evidence that, in the population of teenagers this sample represents, more daily screen time is associated with a decrease in average nightly sleep duration. Because the screen time data were observed rather than randomly assigned, this conclusion is about association, not causation; some other factor could plausibly be driving both variables at once.

- Test statistic: $t = \dfrac{b_1 - 0}{s_{b_1}}$, with $df = n - 2$, using the value of β_1 specified in H_0 (usually 0).
- Standard computer output typically reports a two-sided p-value by default, regardless of which alternative you are actually testing.
- For a one-sided H_a, divide the reported two-sided p-value by 2 only if the sign of t matches the direction of H_a; otherwise the true one-sided p-value exceeds 0.5.
- Reject H_0 when $p < \alpha$; state the conclusion in context, and be careful not to claim causation from an observational study.

Computing in Excel

Continuing from the `LINEST` output in Section 2 (cell D2 for b_1, cell D3 for s_{b_1}):

1. Test statistic: in an empty cell, enter `=D2/D3`. Excel returns -4.0131, matching the T column of the regression output.
2. One-sided p-value directly: enter `=T.DIST(D2/D3,6,TRUE)`. This gives the left-tail probability $P(T < t)$ for $df = 6$, which is exactly the one-sided p-value for $H_a : \beta_1 < 0$ when t is negative. Excel returns 0.0035, with no separate halving step needed.
3. If you only have the two-sided p-value from a printed output table, use `=T.DIST.2T(ABS(D2/D3),6)/2` instead, which reproduces the halving rule from Step 3 directly. Both approaches agree here: 0.0035.

`T.DIST` assumes a lower-tail test by default; for $H_a : \beta_1 > 0$, use `=1-T.DIST(t,df,TRUE)` or the equivalent `=T.DIST.RT(t,df)` instead, since you need the upper tail.

Computing in Python

The full test, from raw data to conclusion, takes only a few lines with `scipy.stats`:

```
import numpy as np
from scipy import stats

```

```
x = np.array([2, 3, 4, 5, 6, 7, 8, 9])
y = np.array([7.85, 8.05, 7.40, 7.55, 6.90, 7.35, 6.70, 6.95])

result = stats.linregress(x, y)
df = len(x) - 2
t_stat = result.slope / result.stderr
p_one_sided = stats.t.cdf(t_stat, df)   # left tail, for Ha: beta1 < 0

print(f"b1 = {result.slope:.4f}, SE(b1) = {result.stderr:.4f}")
print(f"t = {t_stat:.4f}, df = {df}")
print(f"one-sided p-value = {p_one_sided:.4f}")
```

This prints `b1 = -0.1649, SE(b1) = 0.0411`, `t = -4.0131, df = 6`, and `one-sided p-value = 0.0035`, matching the by-hand and Excel results exactly. `stats.t.cdf(t_stat, df)` gives the lower-tail probability directly, which is the correct one-sided p-value whenever H_a is ”less than and t is negative; for a ”greater than” alternative, use `1 - stats.t.cdf(t_stat, df)` instead.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 7)

probability of all 5's $= \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \left(\frac{1}{6}\right)^3 = \frac{1}{216} \approx 0.0046$

Answer to Exercise 1 (on page 7)

probability of at least one heads = 1.0 - probability of all tails $= 1.0 - \left(\frac{1}{2}\right)^5 = 1.0 - \frac{1}{32} = \frac{31}{32} = \approx 0.97$

Answer to Exercise 3 (on page 20)

1. $(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots$, valid for $|x| < 1$

2. $\sqrt{1-2x} = 1 - x - \frac{1}{2}x^2 - \cdots$

Answer to Exercise 4 (on page 21)

$$P(\text{exactly 4 makes}) = \binom{6}{4}(0.6)^4(0.4)^2 = 0.31104$$

Answer to Exercise 5 (on page 32)

$df = n - 1 = 24$. A 95 percent confidence interval splits the remaining 5 percent evenly between the two tails, leaving 2.5 percent, or 0.025, in the upper tail. Reading the t-table at $\mathrm{df} = 24$ and upper-tail area 0.025 gives $t^* = 2.064$.

Answer to Exercise 6 (on page 33)

The $n \geq 30$ threshold is not a hard requirement for using a t-interval. It is one way to satisfy the normality condition when you cannot otherwise verify the shape of the population. If the population itself is known to be normally distributed, or if a dotplot or stemplot of the sample shows no strong skew or outliers, the normality condition is satisfied regardless of how small n is, and a t-interval is entirely appropriate. The student conflated a fallback rule with a strict cutoff.

Answer to Exercise 7 (on page 34)

(a) Matched pairs. The same twenty subscribers are measured twice, so the two sets of measurements are linked subscriber by subscriber, and the natural next step is to subtract to get one sample of twenty differences.

(b) Two independent samples. The comedy-watching group and the drama-watching group are made up of entirely different subscribers with no pairing between them, so there is no meaningful way to subtract one group's values from the other's.

Answer to Exercise 8 (on page 35)

$\mathrm{df} = 15$. A 98 percent confidence interval leaves 1 percent, or 0.01, in each tail, giving $t^* = 2.602$. The standard error is $\dfrac{6.2}{\sqrt{16}} = 1.55$, so the margin of error is $2.602 \times 1.55 \approx 4.03$ minutes. The interval is $28.4 \pm 4.03 \Rightarrow (24.37,\ 32.43)$ minutes.

Answer to Exercise 9 (on page 37)

This is matched pairs, so the sample is the twelve differences, and everything proceeds exactly as it did for a single mean. $\mathrm{df} = 11$, and a 90 percent confidence interval leaves 5 percent in each tail, giving $t^* = 1.796$. The standard error is $\dfrac{4.5}{\sqrt{12}} \approx 1.299$, so the margin of error is $1.796 \times 1.299 \approx 2.33$ minutes. The interval is $-3.2 \pm 2.33 \Rightarrow (-5.53,\ -0.87)$ minutes. Because the entire interval is negative, and the differences were defined as before minus after, this suggests call-handling time decreased after training, by somewhere between about 0.87 and 5.53 minutes on average.

Answer to Exercise 10 (on page 37)

The true mean μ is a fixed, though unknown, number. It is not a random variable, so it makes no sense to assign it a probability of falling anywhere. What is random is the interval itself, since a different sample would have produced a different $\bar{x}$ and a different interval. The corrected interpretation is that the analyst is 95 percent confident that the true mean watch time per session falls between 43.75 and 51.85 minutes, meaning that if this sampling process were repeated many times, about 95 percent of the resulting intervals would capture the true mean.

Answer to Exercise 11 (on page 39)

$H_0 : \mu = 500$, $H_a : \mu \neq 500$. Conditions are satisfied, so use a t-test with $df = 15$. The standard error is $\dfrac{6.4}{\sqrt{16}} = 1.6$, giving $t = \dfrac{498.2 - 500}{1.6} \approx -1.125$. For a two-sided test with $df = 15$, this gives a p-value of about 0.278. Because $0.278 > 0.05$, fail to reject H_0. There is not sufficient evidence that the true mean loaf weight differs from 500 grams.

Answer to Exercise 12 (on page 39)

Since the differences are defined as before minus after, a decrease in heart rate shows up as a positive mean difference, so $H_0 : \mu_d = 0$, $H_a : \mu_d > 0$. This is matched pairs, so $df = 14$. The standard error is $\dfrac{6.1}{\sqrt{15}} \approx 1.575$, giving $t = \dfrac{4.8}{1.575} \approx 3.048$, with a one-sided p-value of about 0.0043. Because $0.0043 < 0.05$, reject H_0. There is sufficient evidence that the regimen decreased mean resting heart rate.

Answer to Exercise 13 (on page 40)

Hypotheses are statements about the population parameter, μ, never about the sample statistic, $\bar{x}$. The sample mean is a single computed number from one sample; it is not something you test a hypothesis about, since it is already known exactly once the sample is collected. The correct hypotheses are $H_0 : \mu = 500$ and $H_a : \mu \neq 500$.

Answer to Exercise 14 (on page 41)

$SE = \sqrt{\dfrac{1.8^2}{15} + \dfrac{2.1^2}{17}} \approx 0.690$. With $df = 29$, $t^* \approx 1.699$, so the margin of error is $1.699 \times$

$0.690 \approx 1.17$ hours. The point estimate is $11.2 - 9.8 = 1.4$ hours, giving $1.4 \pm 1.17 \Rightarrow (0.23,\ 2.57)$ hours.

Answer to Exercise ?? (on page 41)

No. The interval $(0.23, 2.57)$ does contain values above 1, but it also contains values below 1, so it does not provide convincing evidence that the difference exceeds one full hour specifically. The interval does support the weaker claim that Model A's mean battery life is greater than Model B's, since the entire interval is positive, just not the stronger one-hour claim.

Answer to Exercise 16 (on page 42)

Pooling assumes the two population standard deviations are equal, $\sigma_1 = \sigma_2$. Here, the two samples show clearly different spreads, which is evidence against that assumption, not for it. Even setting that aside, the current AP Statistics CED does not include the pooled, equal-variances procedure at all, only the unpooled two-sample t-interval using $\sqrt{s_1^2/n_1 + s_2^2/n_2}$ with technology-computed degrees of freedom. The student should use the unpooled formula regardless of how similar or different the spreads look.

Answer to Exercise 17 (on page 43)

$H_0 : \mu_1 - \mu_2 = 0$, $H_a : \mu_1 - \mu_2 < 0$. The standard error is $\sqrt{\dfrac{2.1^2}{22} + \dfrac{2.8^2}{25}} \approx 0.717$, so $t = \dfrac{8.3 - 9.6}{0.717} \approx -1.813$, with $df \approx 44$. The one-sided p-value is about 0.0383. Since $0.0383 < 0.05$, reject H_0. There is convincing evidence that Store 1's mean wait time is shorter than Store 2's.

Answer to Exercise 18 (on page 45)

This would be a Type I error. The researcher rejected H_0 (concluding Store 1's mean wait time is shorter), but if the true means are actually equal, H_0 was in fact true, so rejecting it was a mistake. Because the test used $\alpha = 0.05$, this kind of error would occur no more than about 5 percent of the time across many repeated studies like this one.

Answer to Exercise 19 (on page 45)

(a) Quantitative, paired. The same 30 members are measured twice, before and after, so the natural approach is to subtract and work with the sample of differences.

(b) Categorical. Proportions of an email being opened or not are categorical counts, not measurements, so this belongs to the tools from *Inference for Proportions*.

(c) Quantitative, one sample. A single random sample is used to estimate one population mean, with no comparison to a second group or a second measurement per subject.

Answer to Exercise 20 (on page 49)

1. X is the amount of fertilizer applied (grams per plant); Y is the resulting plant height (centimeters).
2. $Y = \beta_0 + \beta_1 X + \varepsilon$, where β_0 is the population intercept, β_1 is the population slope relating fertilizer amount to height, and ε is random error.
3. If $\beta_1 = 0$, then fertilizer amount has no linear relationship with plant height in the population; the population regression line is flat, and knowing how much fertilizer a plant received tells you nothing about how tall it will grow.
4. β_1 describes the relationship for the entire population of plants the horticulturist could ever grow, under all possible conditions, not just the ones she happened to collect data from. Even collecting data from every plant she currently owns would still leave β_1 unknown, since her plants are themselves a sample of all plants that could exist under those growing conditions. β_1 is a parameter; only a b_1 calculated from data is ever directly observed.

Answer to Exercise 22 (on page 55)

1. Equal variance. The spread of the residuals is clearly not constant across values of X; it grows as car weight increases.
2. The standard error formula used to build s_{b_1} assumes the same amount of scatter around the line at every value of X. If the scatter actually changes with X, as it does here, the standard error is not accurately estimated, so the resulting confidence interval and p-value cannot be trusted, even if every calculation is performed correctly.

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